The requirement was straightforward: the log analysis tool needed to gather system logs from all devices across the network, store them in a database, analyse them, and take necessary actions.

This NMS was intended for use by large service providers. In the US market, the adoption of GSM mobile phones was slower compared to VoIP (Voice over IP). While mobile telephony was just beginning in India, offices were transitioning from PSTN technology to VoIP, and VoIP phones were becoming commonplace on every employee's desk. Each time a call was made or received, or a VoIP phone was powered on or off, a system log was generated. Our log analysis tool needed to collect and analyse all these logs.

Due to their lightweight nature, network devices, including VoIP phones, used to transmit logs to the NMS station using the UDP port. As you're aware, UDP, or User Datagram Protocol, is an unreliable protocol. Unless the datagram is immediately read, it is lost forever.

The log analysis tool I inherited from the previous release could process approximately 10 logs per second. Essentially, it was a Java application with a single thread that sequentially read, stored, analysed a log, executed configured actions, and then moved on to the next log message from the UDP port. Figure 2 describes the architecture in detail.

In the initial step, the thread retrieves the next available log from the UDP port.

Following normalisation, filters are applied to determine if the message can be disregarded. If the log is deemed unimportant according to the filtering rules, the thread returns to step 1.

In the fourth step, the log message is stored in the database. These stored logs serve as valuable resources for generating reports and conducting further analysis later.

Next, in the fifth step, the thread assesses if the log warrants any actions. The administrator defines actions, such as sending emails to administrators and paging an operator, based on the nature of the log. Subsequently, the thread returns to step 1.

When networks are smaller and the number of generated logs is limited, handling 10 logs per second is usually adequate. However, with the proliferation of VoIP phones, the rate of log generation increased, and our log analysis tool struggled to keep up. Consequently, many logs were dropped at the UDP port. If a dropped log message was critical, the NMS administrator remained unaware of the issue, leading to unreported and unattended network glitches.

We were tasked with improving the performance of the log analysis tool

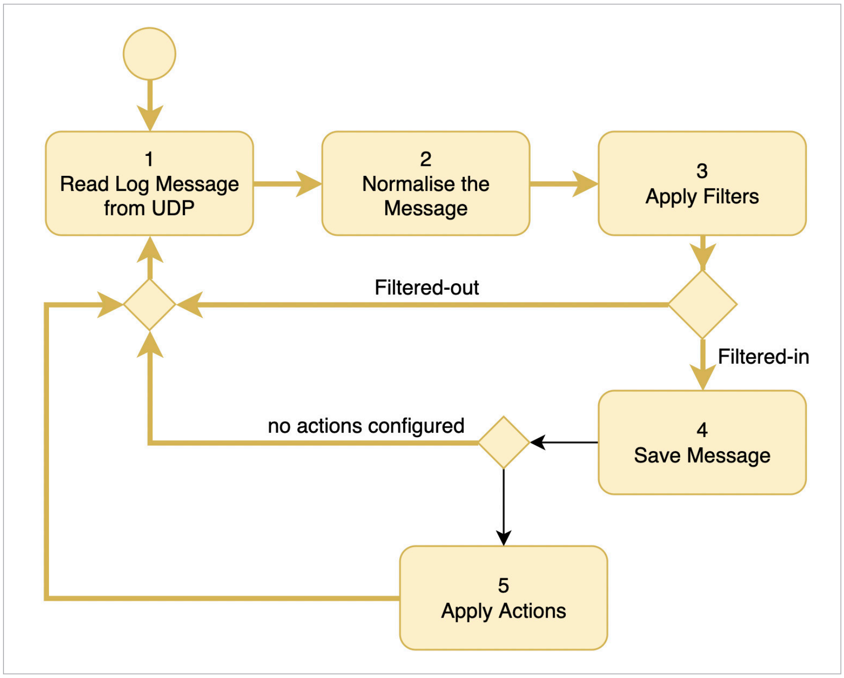

Figure 2: Single-threaded log analyser

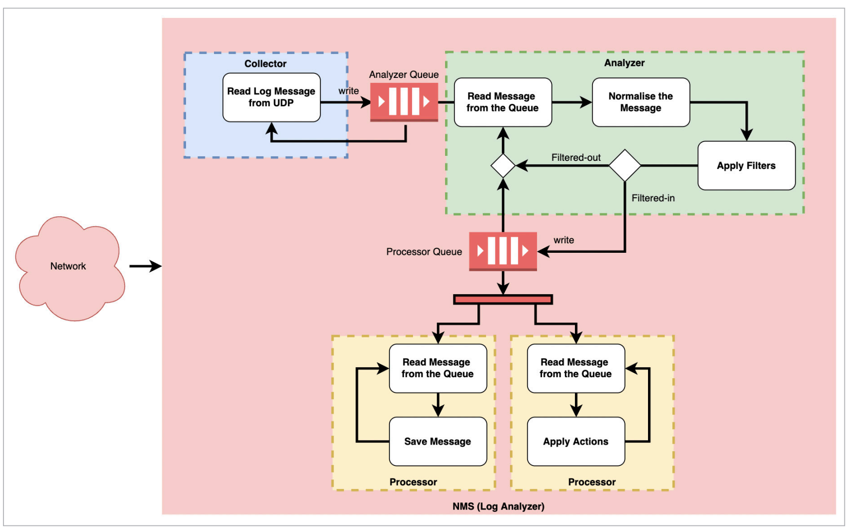

Figure 3: Multi-threaded log analyser